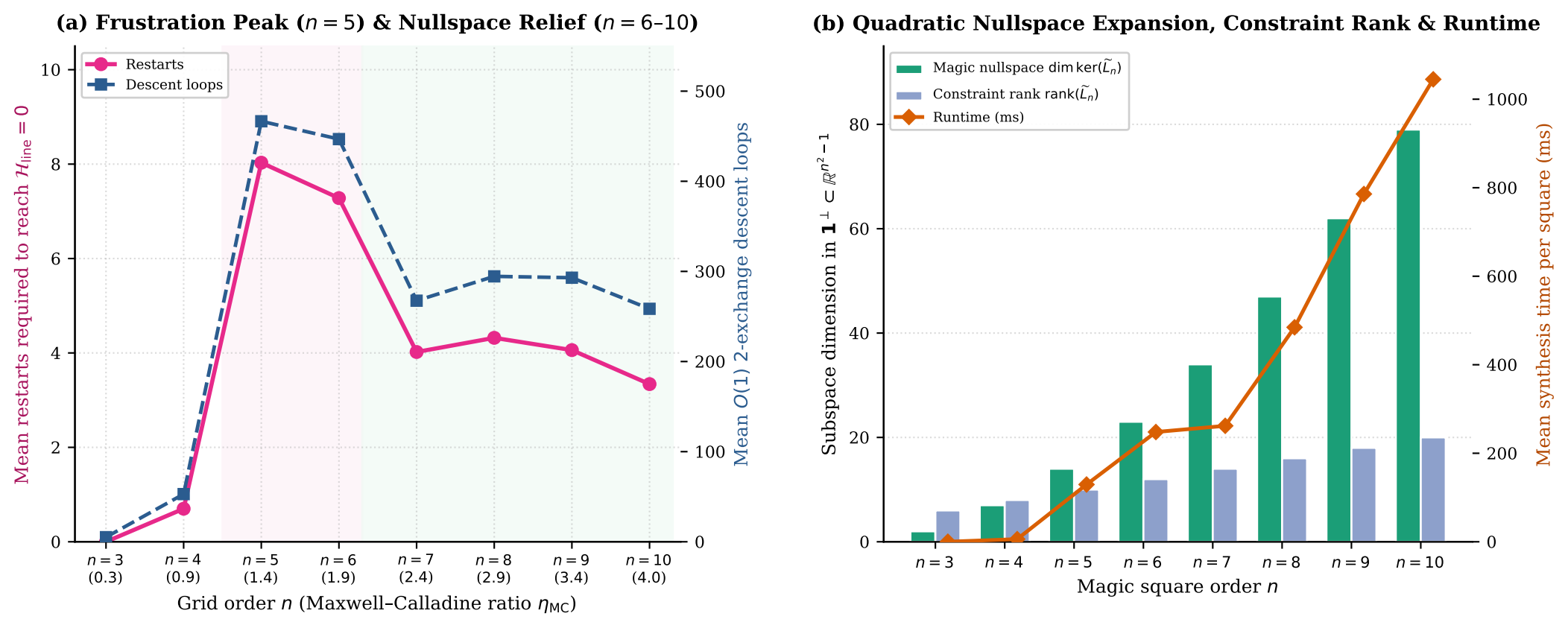

(a) Frustration Peak ($n = 5$) & Nullspace Relief ($n = 6$–10)
Restarts
Descent loops
Mean restarts required to reach $\mathcal{H}_{\text{line}} = 0$
Mean $O(1)$ 2-exchange descent loops
0
2
4
6
8
10
0
100
200
300
400
500
$n = 3$ (0.3)
$n = 4$ (0.9)
$n = 5$ (1.4)
$n = 6$ (1.9)
$n = 7$ (2.4)
$n = 8$ (2.9)
$n = 9$ (3.4)
$n = 10$ (4.0)
Grid order n (Maxwell–Calladine ratio η_{MC})
(b) Quadratic Nullspace Expansion, Constraint Rank & Runtime
Magic nullspace dim $\ker(\widetilde{L}_n)$
Constraint rank $\operatorname{rank}(\widetilde{L}_n)$
Runtime (ms)
Subspace dimension in $\mathbf{1}^{\perp} \subset \mathbb{R}^{n^2-1}$
Mean synthesis time per square (ms)
0
20
40
60
80
0
200
400
600
800
1000
$n = 3$
$n = 4$
$n = 5$
$n = 6$
$n = 7$
$n = 8$
$n = 9$
$n = 10$
Magic square order n